FIFA WORLD CUP

ANALYSIS DASHBOARD

Power BI Project Presentation

Data-Driven Football Intelligence

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AGENDA

What We Will Cover

01 **Project Overview**
Objectives, data sources, data model structure

02 **Data Model & Measures**
Tables, relationships & DAX measures built

03 **Dashboard Pages**
All 7 analysis pages explained in detail

04 **Stadium Analysis**
Capacity, matches hosted, utilization scatter

05 **Teams & Match Analysis**
Wins, losses, draws, goal differential

06 **Goals & Players Analysis**
Goal patterns, top scorers, cards summary

07 **Key Insights & Conclusion**
Main findings from the analysis

PROJECT OVERVIEW

FIFA World Cup Analysis — Scope & Objectives

🎯 PROJECT OBJECTIVE

Build a comprehensive, interactive Power BI dashboard to analyze FIFA World Cup data across multiple dimensions including stadiums, teams, matches, goals, and player performance.

The dashboard enables stakeholders to explore historical tournament data and derive actionable insights through rich visualizations and dynamic filters.

7 — **Dashboard Pages**
Analysis Sections

10+ — **Visual Types**
Chart Varieties

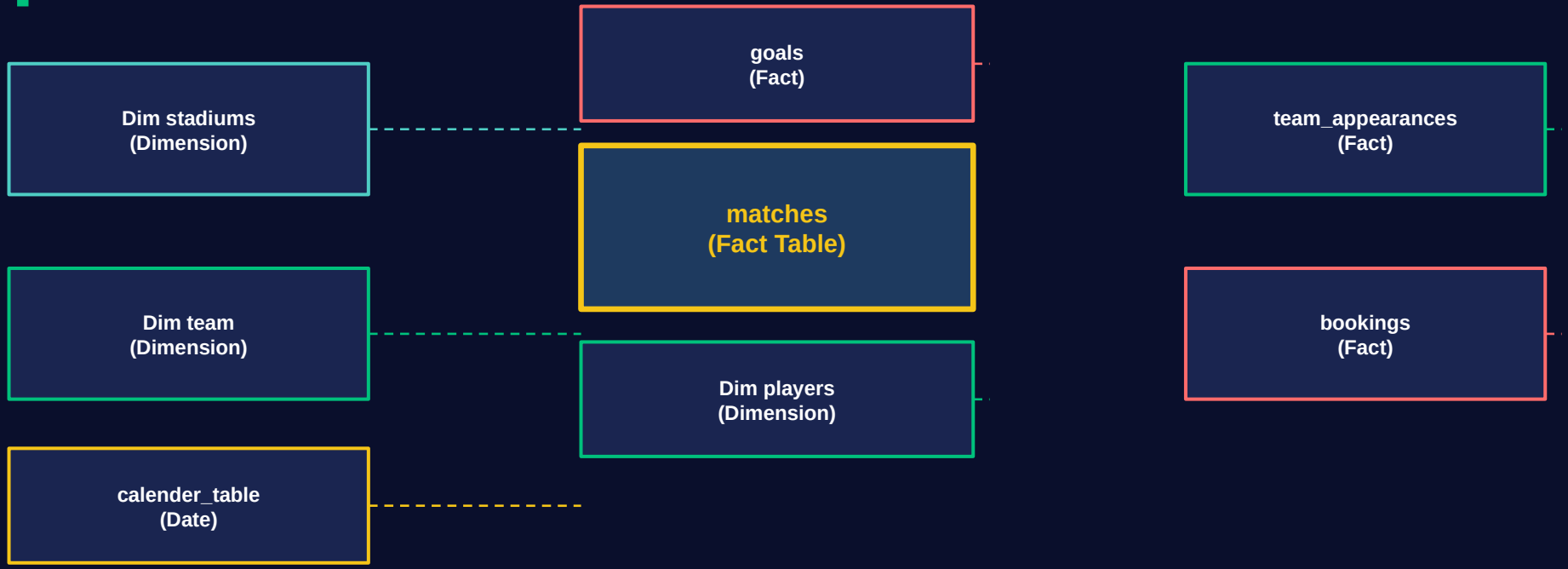
9 — **Data Tables**
Fact & Dimension

📦 DATA SOURCES & TABLES

_Measures Measures Table	**Dim stadiums** Dimension	**Dim team** Dimension	**Dim players** Dimension	**goals** Fact
matches Fact	**team_appearances** Fact	**bookings** Fact	**calender_table** Date Table	

DATA MODEL & DAX MEASURES

Tables, Relationships & Calculated Measures



📐 DAX MEASURES BUILT

Total_Stadium

```
DISTINCTCOUNT(Dim_stadiums[stadium_id])
```

Matches Hosted

```
COUNTROWS(matches)
```

Wins

```
SUM(team_appearances[win])
```

DASHBOARD PAGES

7 Interactive Analysis Sections

01 Overview

KPI cards for Total Teams, Matches, Goals & Stadiums. Line chart of matches by year, treemap of matches by country, donut by stage.

📊 Card, Line, Treemap, Donut

02 Stadium Analysis

Bar chart of matches by stadium, column chart of stadium capacity, scatter plot linking capacity to matches hosted.

📊 Bar, Column, Scatter + Slicers

03 Geographic Distribution

Treemap and bar chart showing matches distribution by country and city, filtered by country & city slicers.

📊 Treemap, Bar + Slicers

04 Teams Performance

Win/Loss/Draw clustered bar, goal differential column chart, total matches bar, wins line chart, and detailed table.

📊 Bar, Column, Line, Table + Slicers

05 Match Analysis

Win/Lose/Draw donut, goals by stage column, goals by minute line chart, matches by stadium bar.

📊 Donut, Column, Line, Bar

06 Goals Analysis

Goals by stage and match period column charts, own goals vs penalty pie chart, bookings table (yellow/red cards).

📊 Column ×2, Pie, Table

07 Players Analysis

Top scorers bar chart, comprehensive player stats table showing goals, matches, yellow & red cards, with slicers.

📊 Bar, Table + Slicers

STADIUM ANALYSIS

Page 2 — Venue Performance & Capacity Intelligence

Total Matches By Stadium

Clustered Bar Chart

`Fields: Dim stadiums.stadium_name → _Measures.Total_Matches`

💡 Reveals which venues hosted the most matches. Identifies the most utilized stadiums throughout the tournament history.

Capacity By Stadium

Clustered Column Chart

`Fields: Dim stadiums.stadium_name → stadium_capacity`

💡 Compares physical capacity across all World Cup venues — crucial for understanding infrastructure scale across different host nations.

Matches vs Capacity Scatter

Scatter Chart

`Fields: stadium_capacity (X) × Total_Matches (Y) + stadium_name`

💡 Uncovers the correlation between a stadium's size and how many games it hosted — larger capacity ≠ always more games.

🔽 Interactive Slicers: Year (calender_table.year) | Stadium Name (Dim stadiums.stadium_name)

TEAMS PERFORMANCE

Page 4 — Win/Loss Analysis, Goal Differential & Rankings

Win / Loss / Draw By Team

Clustered Bar

`Fields: Dim team.team_name → win, lose, draw`

Side-by-side comparison of all three results per team — instantly shows which teams were the most dominant or struggled most.

Goal Differential By Team

Column Chart

`Fields: Dim team.team_name → goal_differential`

Net goals scored minus conceded — a key metric for overall team strength across the tournament.

Total Matches By Team

Bar Chart

`Fields: Dim team.team_name → Total_Matches`

Teams that advanced far naturally hosted more matches — this chart directly tracks participation depth.

Wins By Team (Trend)

Line Chart

`Fields: Dim team.team_name → Wins (measure)`

Shows teams with consistently high win rates, highlighting the elite contenders across all World Cup editions.

Summary Table

The tableEx visual provides a full breakdown per team:

- ✦ Team Name
- ✦ Stage Name
- ✦ Total Matches

Slicers allow filtering by Wins, Losses, Draws and Goal Differential ranges for deep-dive exploration.

🔽 Slicers: Wins | Losses | Draws | Goal Differential

GOALS & MATCH ANALYSIS

Pages 5 & 6 — Match Patterns, Goal Timing & Bookings

PAGE 5 — MATCH ANALYSIS

Donut Chart

Win/Lose/Draw by Result — shows overall result distribution across all tournaments

Column Chart

Total Goals by Stage Name (groups.stage_name) — group vs knockout goals comparison

Line Chart

Total Goals by Minute Label — uncovers peak scoring windows during matches (e.g. 80-90 min)

Bar Chart

Total Matches by Stadium — cross-validates with Stadium Analysis page

PAGE 6 — GOALS ANALYSIS

Column Chart

Goals by Stage — identifies which tournament stages produce the most goals

Column Chart

Goals by Match Period — 1st Half vs 2nd Half vs Extra Time breakdown

Pie Chart

Own Goals vs Penalties — proportion of non-open-play goals in World Cup history

Table (Bookings)

Minute Label × Yellow Cards × Red Cards × 2nd Yellow — disciplinary deep-dive

PLAYERS ANALYSIS

Page 7 — Top Scorers, Appearances & Disciplinary Records

📊 Top Scorers — Bar Chart

Visual: Bar Chart
Fields: Dim players.family_name → _Measures.Total_Goals

This horizontal bar chart ranks all World Cup players by their total goals scored across their entire international career in the tournament.

It allows instant identification of the all-time top scorers and shows the gap between elite goal-scorers and average performers.

📋 Player Stats Table

The tableEx visual provides a comprehensive multi-metric player comparison:

- 👤 Player (Family Name)
- ⚽ Total Matches
- 🎯 Total Goals
- 🟡 Yellow Cards
- 🔴 Red Cards

🔽 Interactive Slicers:

- Red Cards (bookings.red_card)
- Yellow Cards (bookings.yellow_card)
- 2nd Yellow Card
- Match Period (goals.match_period)

KEY INSIGHTS

What the Dashboard Reveals — Top Findings from the Analysis

🏟 Stadiums

A small number of large-capacity stadiums consistently host the most matches, while mid-size venues see far less utilization — capacity alone doesn't determine usage frequency.

🏆 Teams

Brazil, Germany and Argentina dominate match counts and wins. Goal differentials show elite teams not just win but win convincingly — massive spread from top to bottom.

⚽ Goals

Goals spike in the final 15 minutes of both halves. The knockout stages produce more goals per match than group stages. Own goals and penalties form a notable portion of the total.

🌍 Geography

European and South American host nations dominate match counts. Some host cities see far more matches than others, reflecting strategic venue selection by FIFA.

🌟 Players

A very small group of players account for the majority of all World Cup goals. The correlation between matches played and goals scored is strong but not universal — some impact players in few games.

🟡 Discipline

Yellow cards peak in the group stage as teams compete for qualification. Red cards are rare but decisive. Second yellow cards occur most in high-stakes knockout matches.

CONCLUSION

This Power BI dashboard transforms raw FIFA World Cup data into a rich, multi-dimensional analytical experience — covering stadiums, teams, matches, goals, and player performance in one unified, interactive report.

7	9	6+	10+
Dashboard Pages	Data Tables	DAX Measures	Visual Types

Thank You!

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